

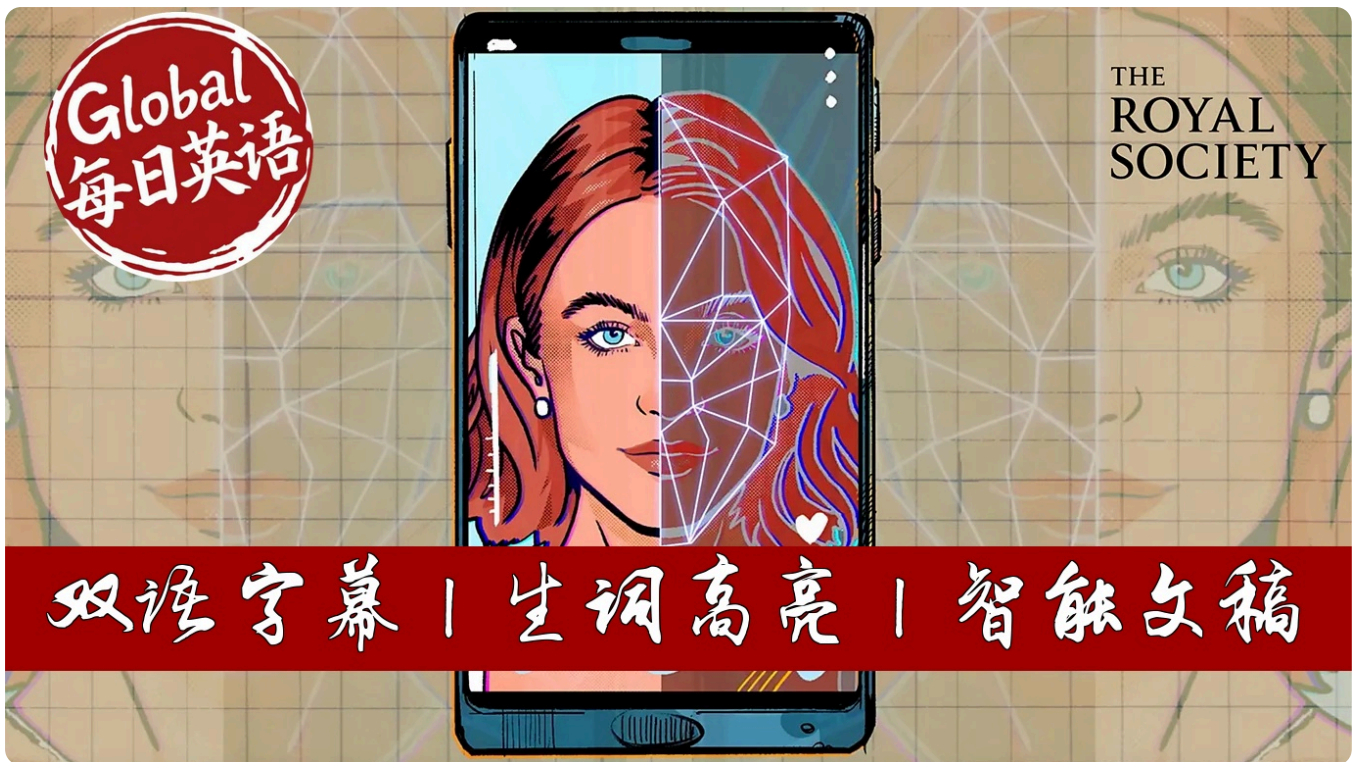
Is there really a mathematical formula for a beautiful face?

【BBC精选：漂亮的脸真的有数学公式吗？】

Summary: The Golden Ratio, a mathematical concept discovered by Fibonacci, has been hailed as the key to beauty in nature, art, and even human faces. It appears in phenomena from tornadoes to sunflowers and inspired historical studies linking it to aesthetic preferences. However, modern research debunks many of these claims, showing that the preference for golden rectangles may be biased and that natural spirals don't always follow the ratio. Despite its appeal in popular culture and cosmetic trends, the golden ratio's connection to beauty is largely unproven. Ultimately, beauty remains subjective, beyond any mathematical formula.

摘要：黄金比例这一数学概念由斐波那契发现，曾被推崇为自然、艺术乃至人脸美的关键。它出现在从龙卷风到向日葵等各种现象中，并引发历史上将其与审美偏好相联系的研究。然而现代研究驳斥了这些观点，指出对黄金矩形的偏好可能存在偏差，且自然界的螺旋并非都遵循该比例。尽管在流行文化和美容趋势中备受追捧，黄金比例与美学的关联大多缺乏科学依据。最终，美是主观的，超越任何数学公式的界定。





🕒 Estimated Reading Time: 7 min.

📖 四级生词 📖 六级生词 📖 雅思生词 📖 托福生词 📖 专八生词 📖 SAT生词 📖 考研生词 📖 GRE生词

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▶ The Golden Ratio is a mathematical phenomenon that can explain everything from tornadoes to the Mona Lisa,

黄金#比例#是一种#数学#上的#现象#, 据说可以解释从龙卷风到《蒙娜丽莎》的一切,

▶ the pyramids to sunflowers and seashells, and why some people are much more attractive than others,

从金字塔到向日葵和贝壳, 甚至解释为什么有些人比别人更有吸引力,

▶ at least according to thousands of TikTok posts and a bunch of entrepreneurial cosmetic surgeons.

至少根据数千条 TikTok 帖子和一批创业型整形外科医生的说法。

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▶ So how did we come to consider this obscure formula to be both the secret of life and the ultimate standard of beauty?

那么我们是如何将这个#晦涩#的公式视为生命的秘密和终极的美学标准的呢？

▶ Way back in 1202, Leonardo of Pisa, also known as Fibonacci,

早在1202年，比萨的列奥纳多，也就是斐波那契，

▶ wrote a book in which he introduced Europe to the Arabic numerals we use today.

写了一本书，把今天我们使用的阿拉伯数字介绍到欧洲。

▶ But buried in chapter 12, there was a sequence of numbers discovered by the ancient Hindu poet Pingala.

但在第12章中，隐藏着一个由古代印度#诗人#平伽罗发现的数字#序列#。

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▶ It goes something like this.

它大概是这样的：

▶ In this sequence, each new number is the sum of the previous two.

在这个#序列#中，每个新数字都是前两个数字的和。

▶ So the first and second numbers add up to the third.

因此，第一和第二个数相加得到第三个数。

▶ Third and fourth add up to the fifth and so on.

第三和第四相加得到第五个数，依此类推。

▶ Fibonacci explained this progression using the example of a field of incestuous rabbits.

斐波那契用一群近亲繁殖的兔子来解释这种#数列递进#。

▶ Say we want to draw a rectangle.

假设我们要画一个#矩形#。

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▶ Let's make the length one Fibonacci number and the height the next Fibonacci number.

我们用一个斐波那契数作长，另一个相邻的斐波那契数作高。

▶ This is known as the golden rectangle.

这就被称为黄金#矩形#。

▶ The approximate ratio of the height to the width is called the golden ratio or the golden section.

高与宽的近似#比例#被称为黄金#比例#或黄金分割。

▶ The extraordinary thing about this rectangle is that if you cut off a perfect square,

这个#矩形#的奇妙之处在于，如果切去一个正方形，

▶ you're left with a smaller rectangle of exactly the same proportions.

你会得到一个比例完全相同的小#矩形#。

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▶ Now watch what happens when we start cutting out smaller and smaller squares.

现在看看当我们不断切出越来越小的正方形时会发生什么。

▶ In the 1860s, a German psychologist named Gustav Fekner showed people a variety of rectangles and asked them to pick their favourite.

在19世纪60年代，一位名叫古斯塔夫·费希纳的德国#心理学家#展示了各种矩形，让人们挑选他们最喜欢的。

▶ He reported that the winner was the golden rectangle.

他报告说，获胜的是黄金#矩形#。

▶ And so began the beguiling idea that there's a particular proportion that is the most beautiful to the human eye.

于是产生了这样一个诱人的观念：有一种比例是人眼看来最美的。

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▶ And you don't have to look far to see the golden ratio all around us.

你不必费力，就能在我们周围看到黄金#比例#的存在。

▶ In head of some flasseeds, there are two sets of interlocking spirals.

在一些亚麻籽的花盘中，有两组#交错#的螺旋。

▶ If you count all the spirals going one way, you get one Fibonacci number and if you count all the spirals going the other way,

如果你数一个方向的所有螺旋，会得到一个斐波那契数，而数另一个方向的螺旋，

▶ you'll get the next Fibonacci number.

你会得到下一个斐波那契数。

▶ But nature is not always so obliging.

但大自然并不总是如此#顺从#。

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▶ Fails, for example, grow their shells at a rate dependent on how much calcium is available,

例如蜗牛，它们的壳生长速度#取决于#可获得的#钙#含量，

▶ not according to the golden ratio.

而不是按照黄金#比例#。

▶ There are many spirals in nature and some of them are very mathematically interesting.

自然界中有许多螺旋，其中一些在数学上确实非常有趣。

▶ But just because something is a spiral does not mean it's a golden spiral.

但某物是#螺旋#，并不代表它就是黄金#螺旋#。

▶ Associations between the golden ratio and what humans find beautiful may be even more bogus.

黄金#比例#与人类审美的联系可能更为#虚假#。

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▶ Remember Gustav Fekner's study?

还记得费希纳的研究吗？

▶ It was disproved in 1974 when University of Toronto Professor Michael Godkowitz concluded

1974年，多伦多大学教授迈克尔·戈德科维奇推翻了它，他得出结论：

▶ the reported preference for the golden rectangle was due to its position in the range of rectangles.

所谓人们偏爱黄金#矩形#，只是因为它在矩形范围中的位置。

▶ More recent studies have found that, given 48 options,

更近的研究发现，当有48个选项时，

▶ it is almost impossible to even pick out the golden rectangle by eye.

人眼几乎不可能准确挑出黄金#矩形#。

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▶ And despite containing some pretty interesting maths,

尽管帕特农神庙蕴含一些有趣的#数学#，

▶ the parthenon in Athens does not display the golden ratio.

但雅典的帕特农神庙并未体现黄金#比例#。

▶ More concerning are the supposed links between the golden ratio and facial beauty,

更令人担忧的是黄金#比例#与#面部#美感之间的所谓联系，

▶ which over the past decades have seen a slew of inventions.

过去几十年间，这催生了一#大批#发明。

🎵 From a golden mask which was meant to be the blueprint for the perfect face.

从声称是完美面孔#蓝图#的黄金#面具#，

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🎵 To more recent grid filters on TikTok, claiming to test or correct your face according to the golden ratio.

到最近 TikTok 上的#网格#滤镜，声称能根据黄金#比例#测试或修正你的脸型。

🎵 Perpetuating not only unrealistic beauty standards but bad maths too.

这不仅延续了#不切实际#的美貌标准，还传播了错误的#数学#观念。

🎵 The idea that one number can explain everything, from rams, horns and flowers,

一个数字可以解释一切，从公羊、犄角和花朵，

🎵 to artistic masterpieces and human faces is an appealing one.

到艺术杰作和人类面孔，这个想法确实很#诱人#。

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🎵 And speaks to our need to make sense of a chaotic universe.

它反映了人类试图理解混沌宇宙的需求。

🎵 The golden ratio is part of an interwoven universe.

黄金#比例#是宇宙相互交织的一部分。

🎵 Not of ancient art, history and nature, but of mathematics and geometry,

它不仅属于古代艺术、历史与自然，更属于数学与#几何学#，

🎵 an interwovenness that can be actually proven.

这种交织性是可以被真正证明的。

🎵 The good news is there is no mathematical law that defines who is and isn't beautiful.

好消息是，没有任何#数学#定律能定义谁美谁不美。

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 Regardless of the golden ratio, someone somewhere thinks you're a total stunner.

无论黄金#比例#与否，总会有人觉得你惊艳无比。

